Domain grounding and Retrieval Augmented Generation (RAG)

Lab Overview

In this lab, you will build a Retrieval-Augmented Generation (RAG) client application using your own data in Microsoft Foundry (Azure AI Foundry) and run it locally in Visual Studio Code.

You will:

- Create a Foundry hub and project
- Deploy embedding and chat models
- Upload custom PDF data
- Create a vector index using Azure AI Search
- Test grounding in Chat Playground
- Build and run a Python RAG client app in VS Code

What You'll Learn

By completing this lab, you will:

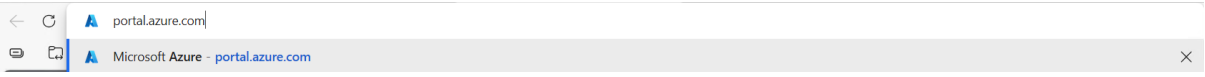
- Deploy Azure OpenAI models
- Create a vector index using Azure AI Search
- Connect a Python client app to Azure OpenAI and Azure AI Search
- Run a grounded AI chatbot locally

Prerequisites

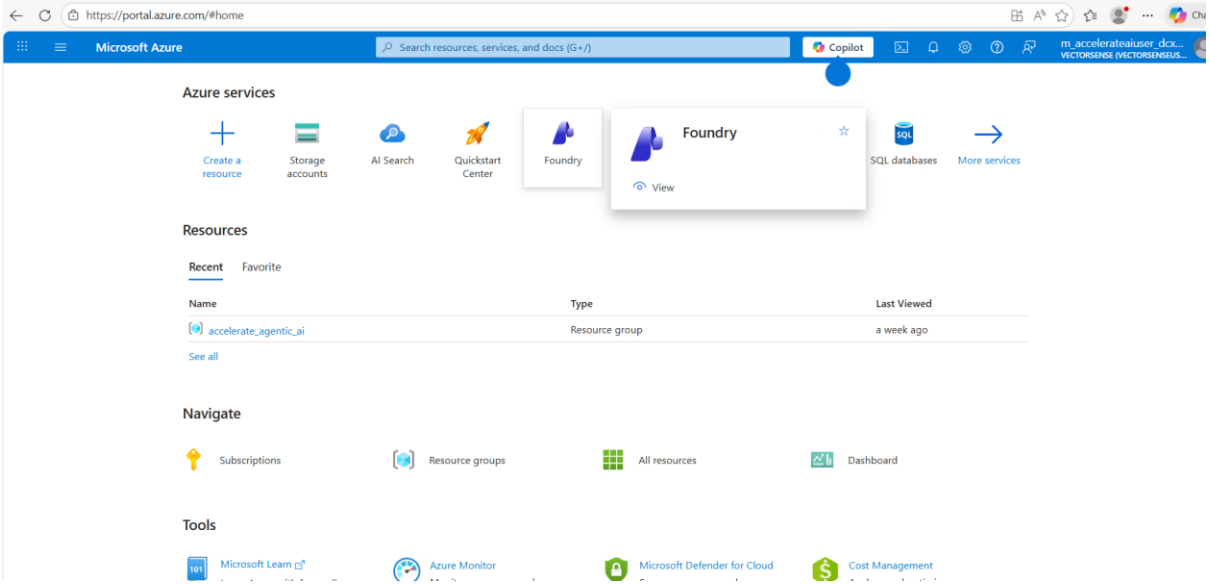
1. Azure subscription
2. Access to https://ai.azure.com
3. Visual Studio Code installed
4. Python 3.10+ installed

Step by step Instructions

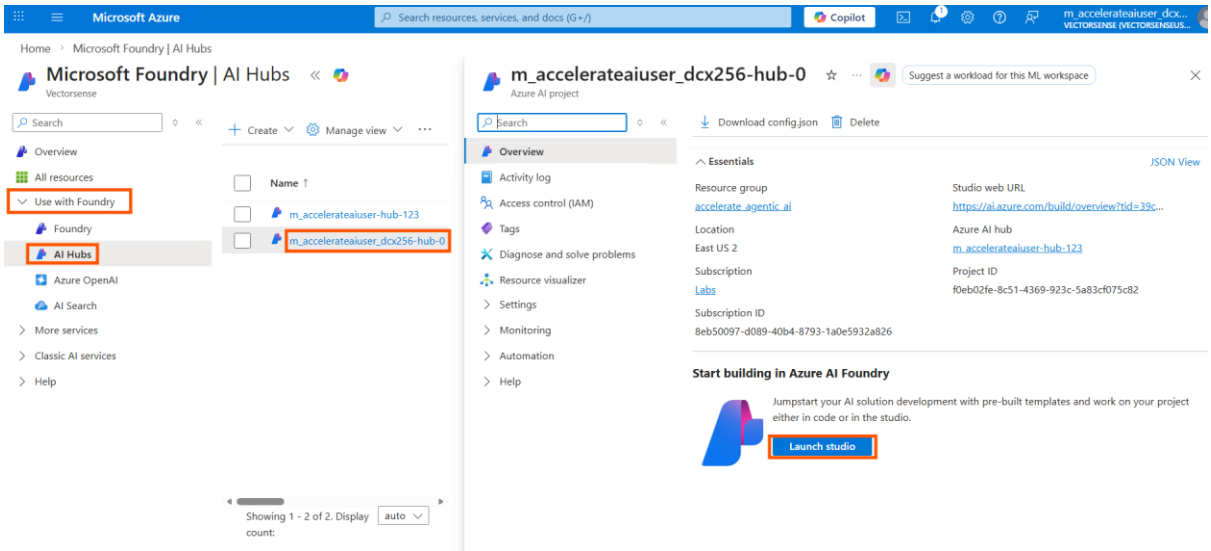
1. Open **Microsoft Edge** and go to https://portal.azure.com. Sign in with the provided credentials.



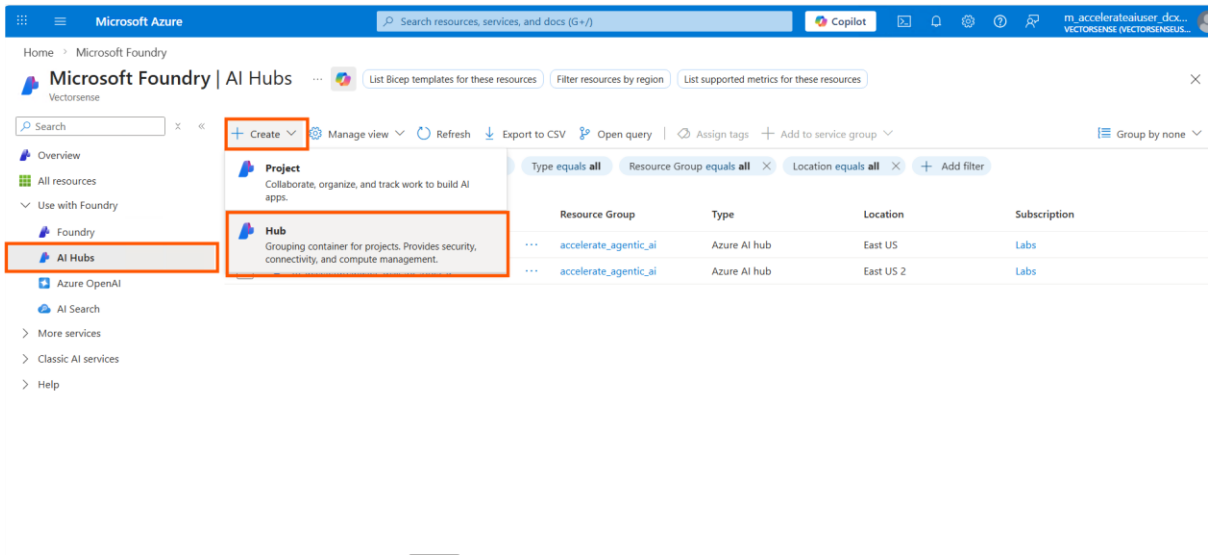
2. Search **Foundry** or click on **Foundry** from the main menu.



3. Click on **AI Hubs** under **Use with Foundry** and click on **Azure AI project** as shown in the image. Click on **Launch studio.**
   <u>We have already deployed the Azure AI Hub and the project so you can directly **jump to step 8**.</u> The below instructions can be used if you cannot find the deployed hub and project.

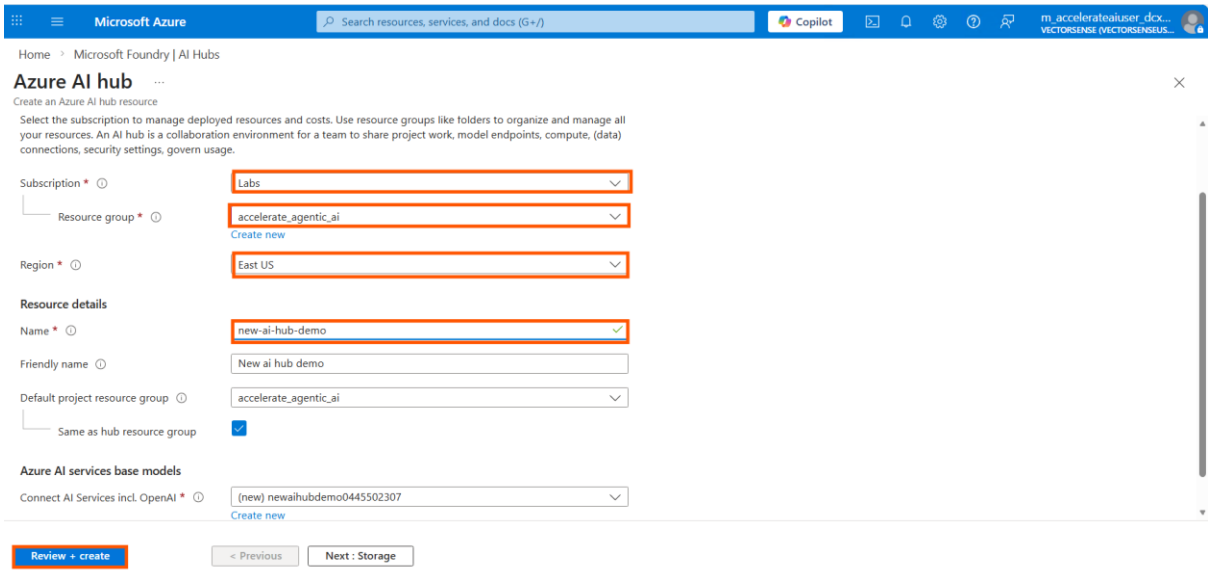


4. Under Microsoft Foundry, click on **AI Hubs -> Create -> Hub.**

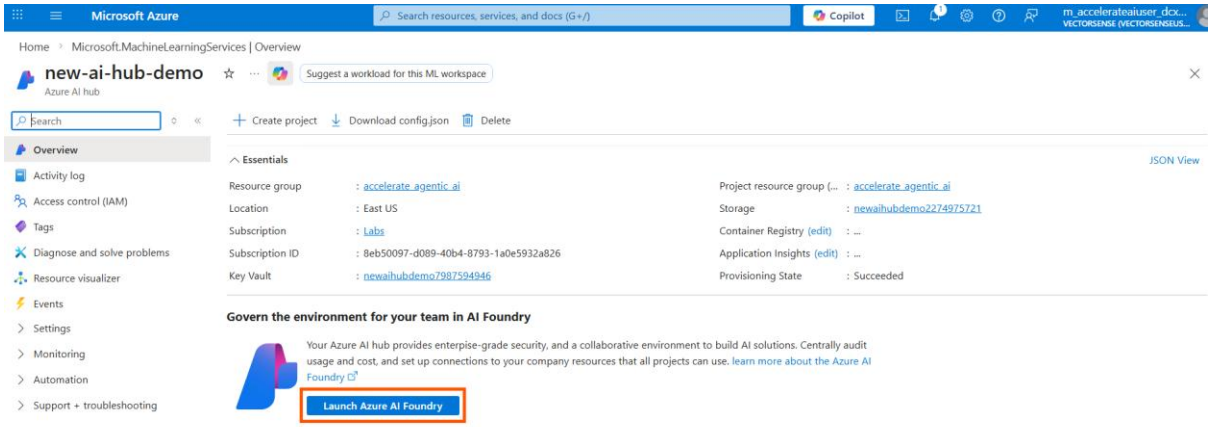


5. Under **Azure AI Hub,** configure. Click on **Review + create** -> **Create**:

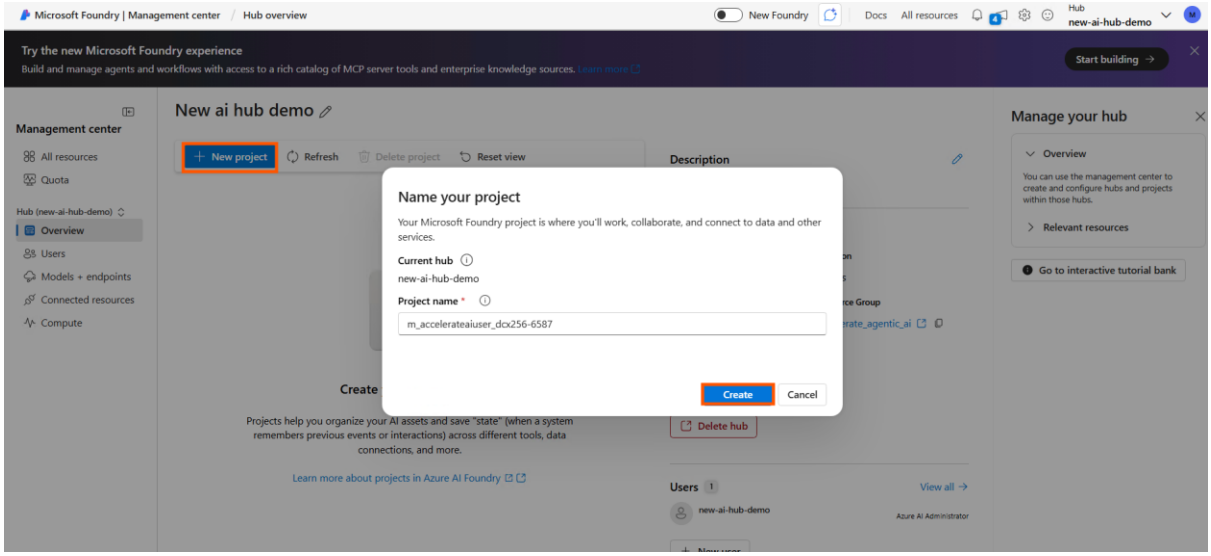
- Subscription: Labs
- Resource Group: Create new or default
- Region: East US/France Central/Sweden Central
- Name: Unique name



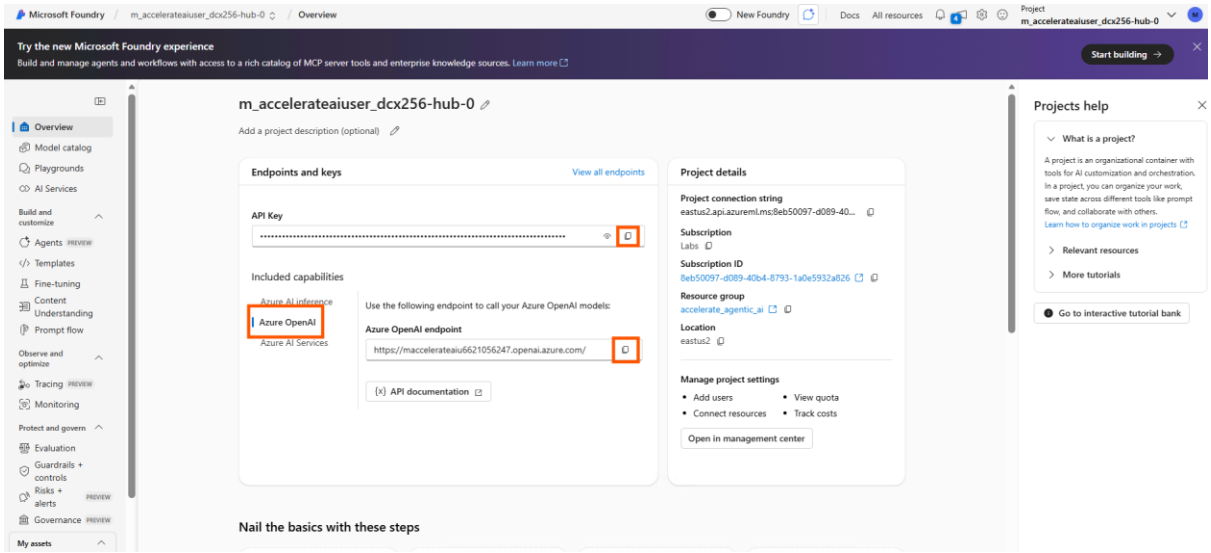
6. Click on **Go to resource -> Launch Azure AI Foundry**



7. Click on **+ New project -> Create**

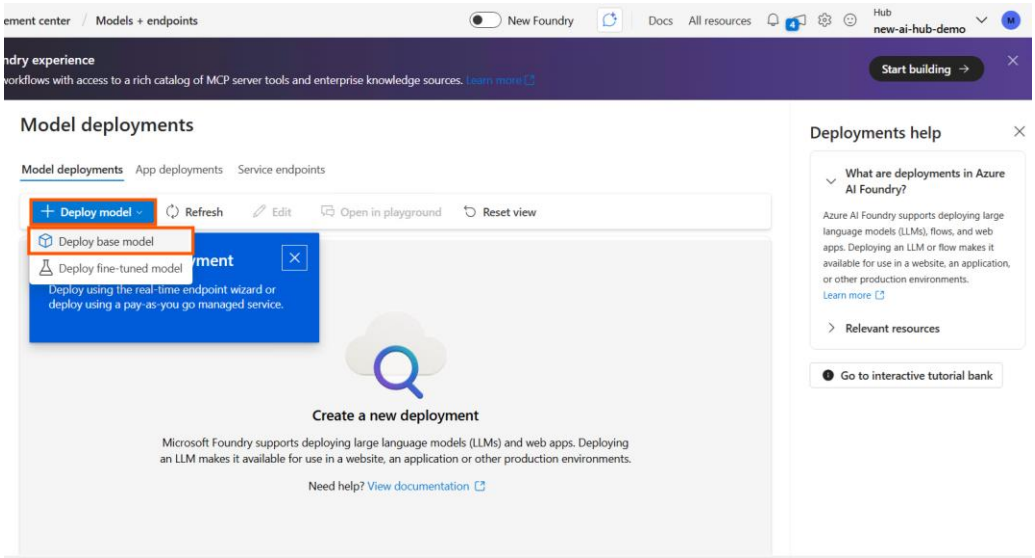


8. **Copy and note** the **API Key** and **Azure OpenAI endpoint** inside notepad

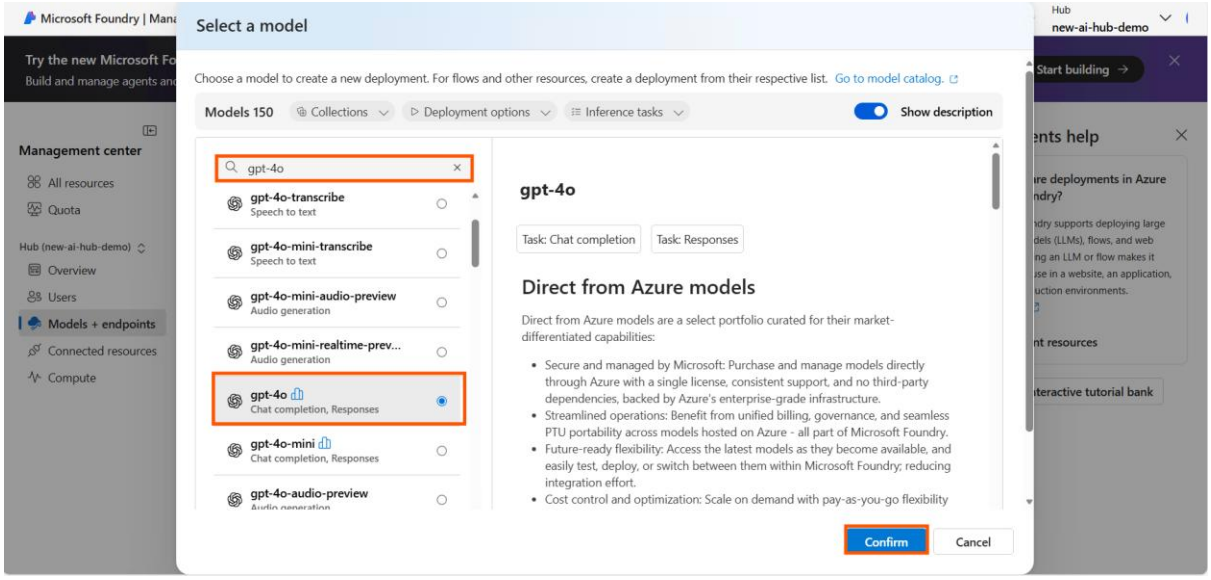


9. <u>We have already deployed the required models, **jump directly to step 13**</u>.
   Below are the instructions if you do not find any deployed model.

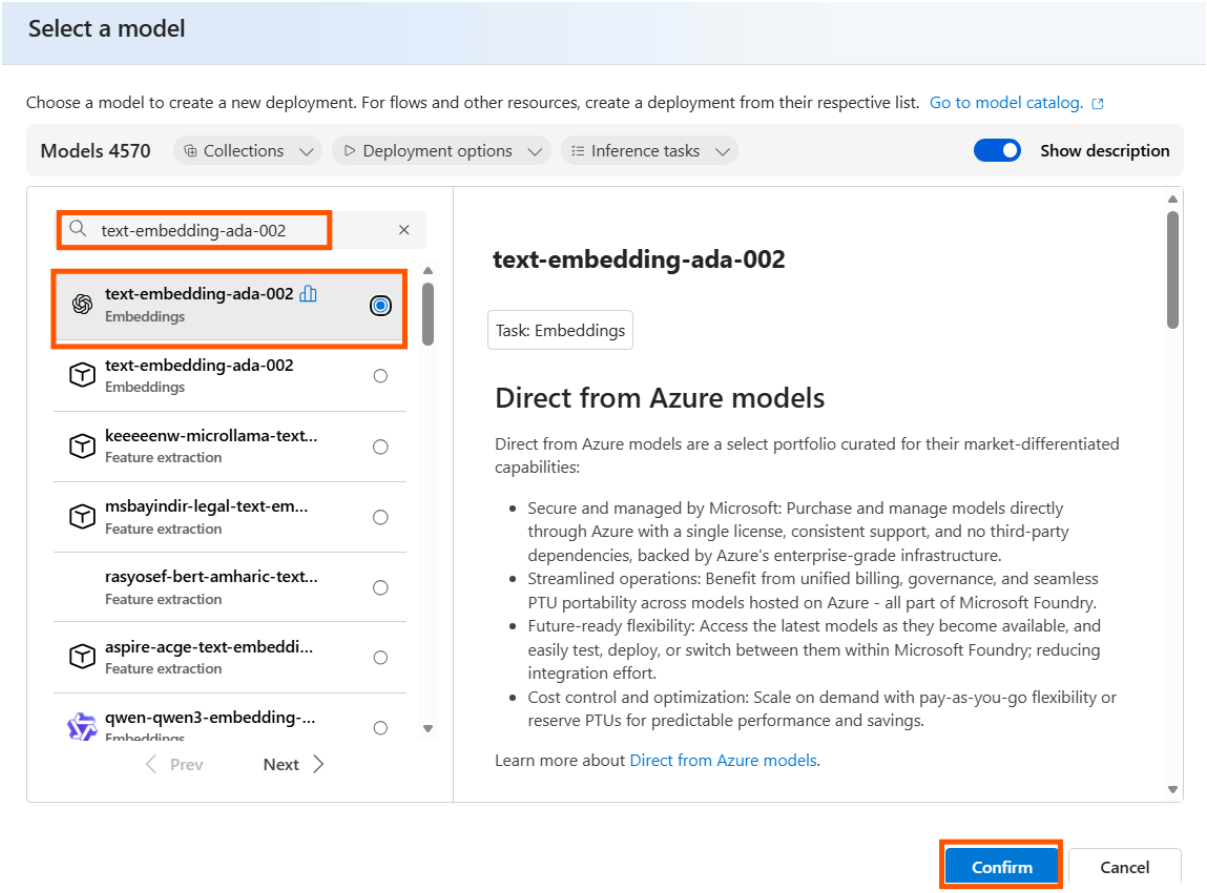
10. Scroll down in the left pane and click on **Models + endpoints -> + Deploy model -> Deploy base model**



11. Search **gpt-4o ->** select **gpt-4o** -> **Confirm -> Connect and deploy.** You might need to change the location to East US/France Central/Sweden Central.



12. Search: text-embedding-ada-002 Search **text-embedding-ada-002 ->** select **text-embedding-ada-002** -> **Confirm -> Deploy**

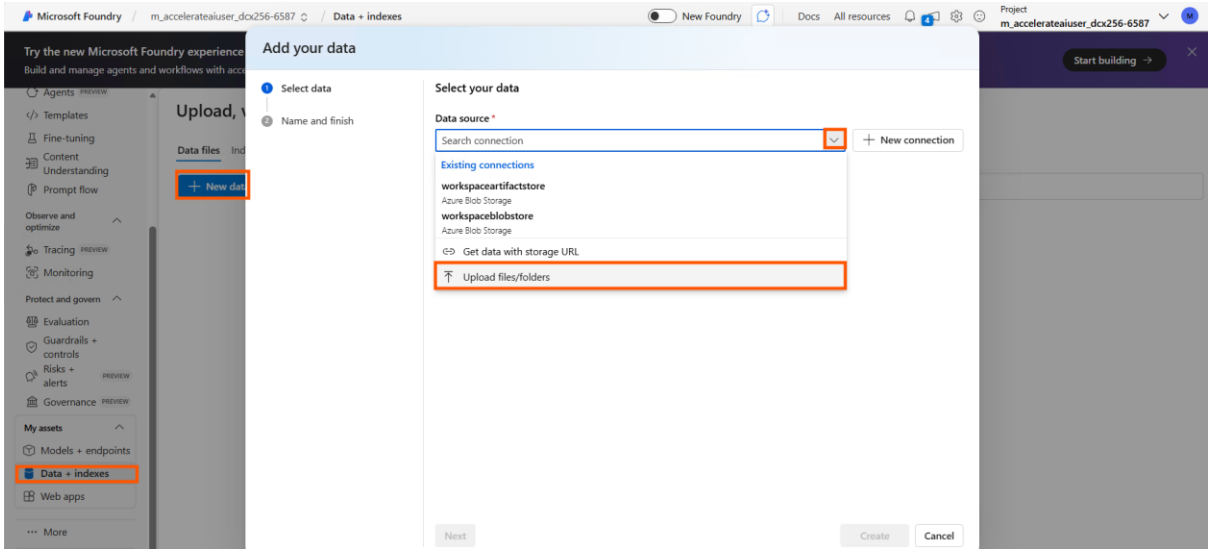


13. We have already deployed the Azure AI Search Index, please **jump directly to step 24** to continue. The below instructions can be used if you don't find any search index.
14. In a new tab, paste the below link and download the lap supporting files:

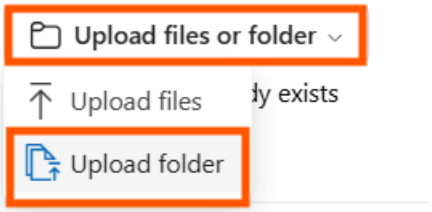
Brochures.zip

(https://vectorsenselabs.blob.core.windows.net/isodata/Accelerate-Agentic_AI/brochures.zip?sp=r&st=2026-02-17T18:50:26Z&se=2027-02-18T03:05:26Z&spr=https&sv=2024-11-04&sr=b&sig=GpkcrrN4JRfXDTgZ7keK4vUmTYgaOHoJ1EGluEvZjdo%3D)

15. After the download completes, close the tab. Open **File Explorer** from the bottom menu bar. Click on **Downloads**. **Right click** over the recently downloaded folder, select **Extract All…** Click **Extract.**
16. Once done, close the File explorer and open Microsoft Edge. Under **My Assets** go to **Data + Indexes -> + New Data -> Dropdown -> Upload file/folders**



17. Click on **Upload files or folder -> Upload folder ->** select **brochures -> Open -> Next**



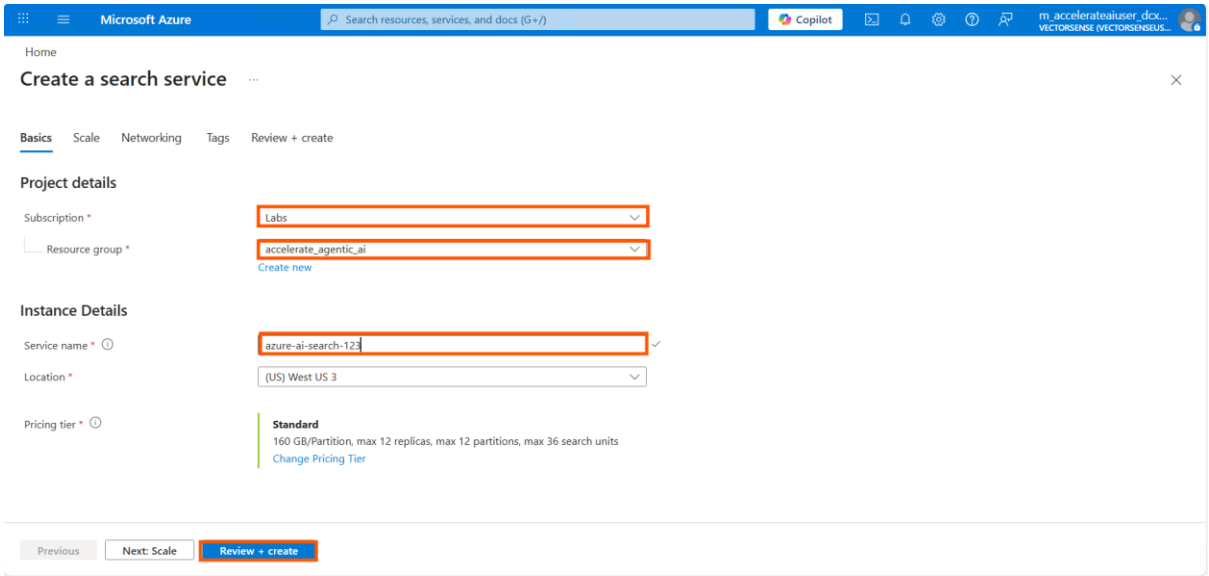
18. Name the data as **brochures -> Create**
19. In the left navigation pane, go to **Data + Indexes -> Indexes -> New Index**

Configure:

Source -> **Data in Foundry** -> select **brochures** -> **Next**

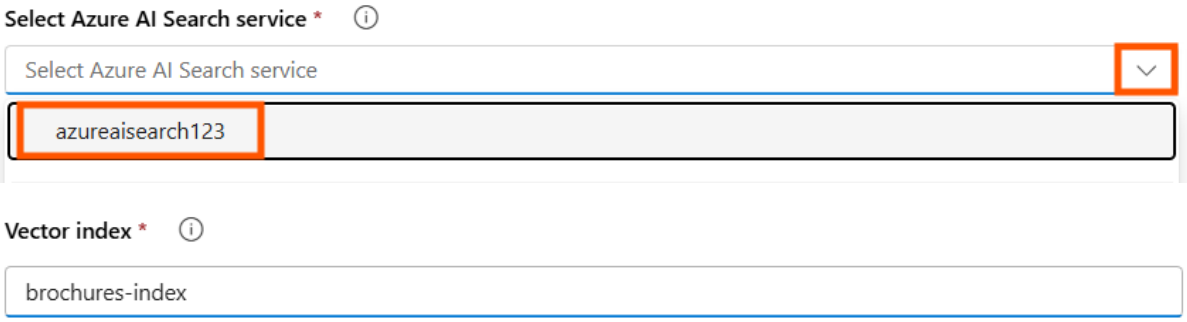
20. Click on **Create a new Azure AI Search resource -> Review + create -> Create -> Go to resource**

- Same subscription
- Same resource group
- Unique service name
- Standard tier
- Default region/East US/France Central/Sweden Central

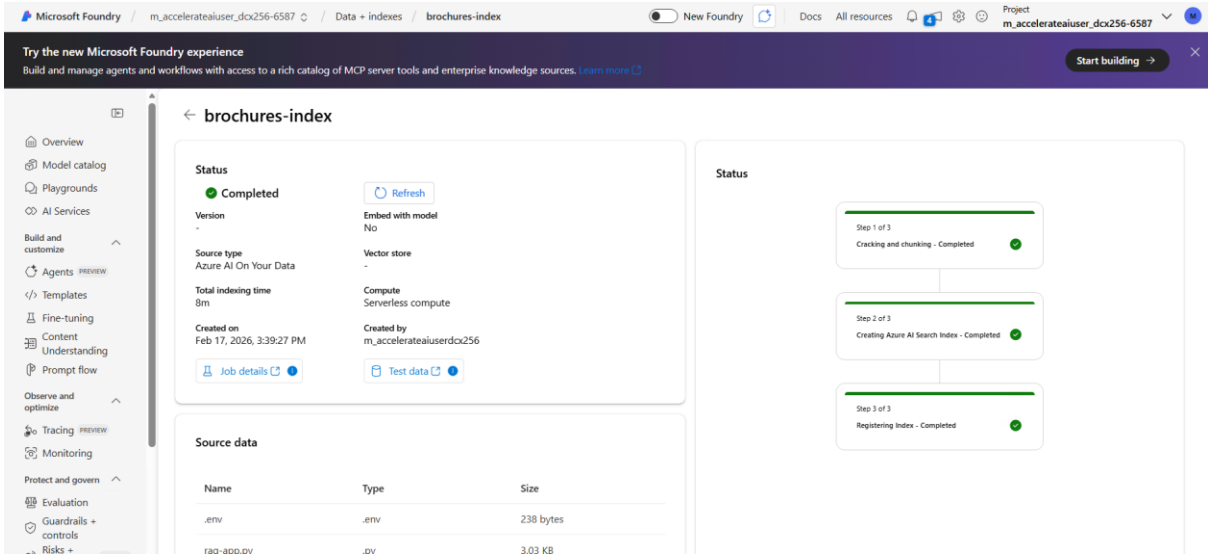


21. Go back to the Foundry tab. Inside the modal click on the dropdown next to Select Azure AI Search service -> Connect other Azure AI Search resource -> Add connection

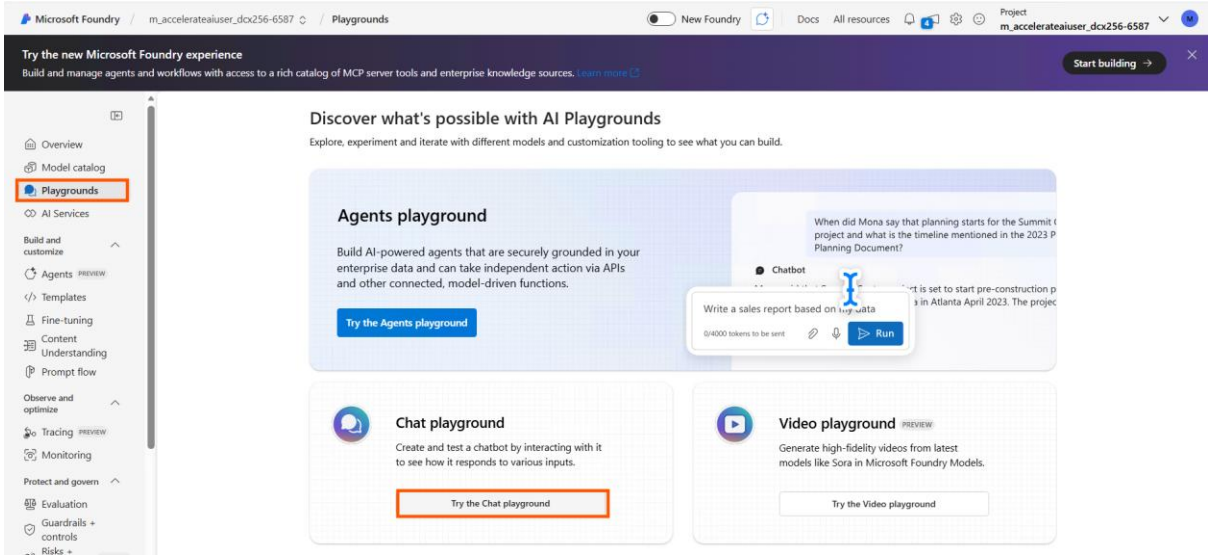
22. Click on the dropdown next to Select Azure AI Search service -> select newly created service -> update Vector index as brochures-index



23. Click Next -> Next -> Create Vector index. Wait for indexing to complete. It might take 5-10 minutes. Watch for the status on the right side, wait until all the 3 steps get executed.



24. In the left navigation pane, go to **Playgrounds → Try the Chat Playground**

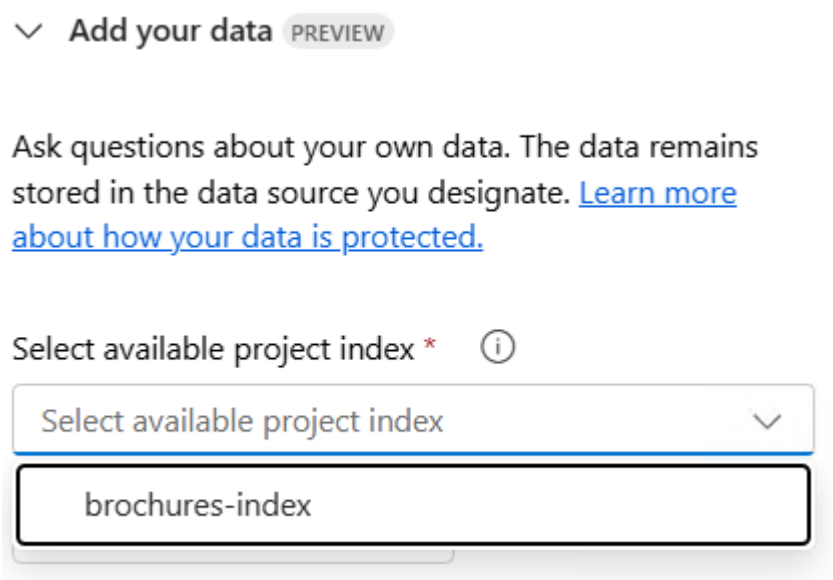


Ask:

Where can I stay in New York?

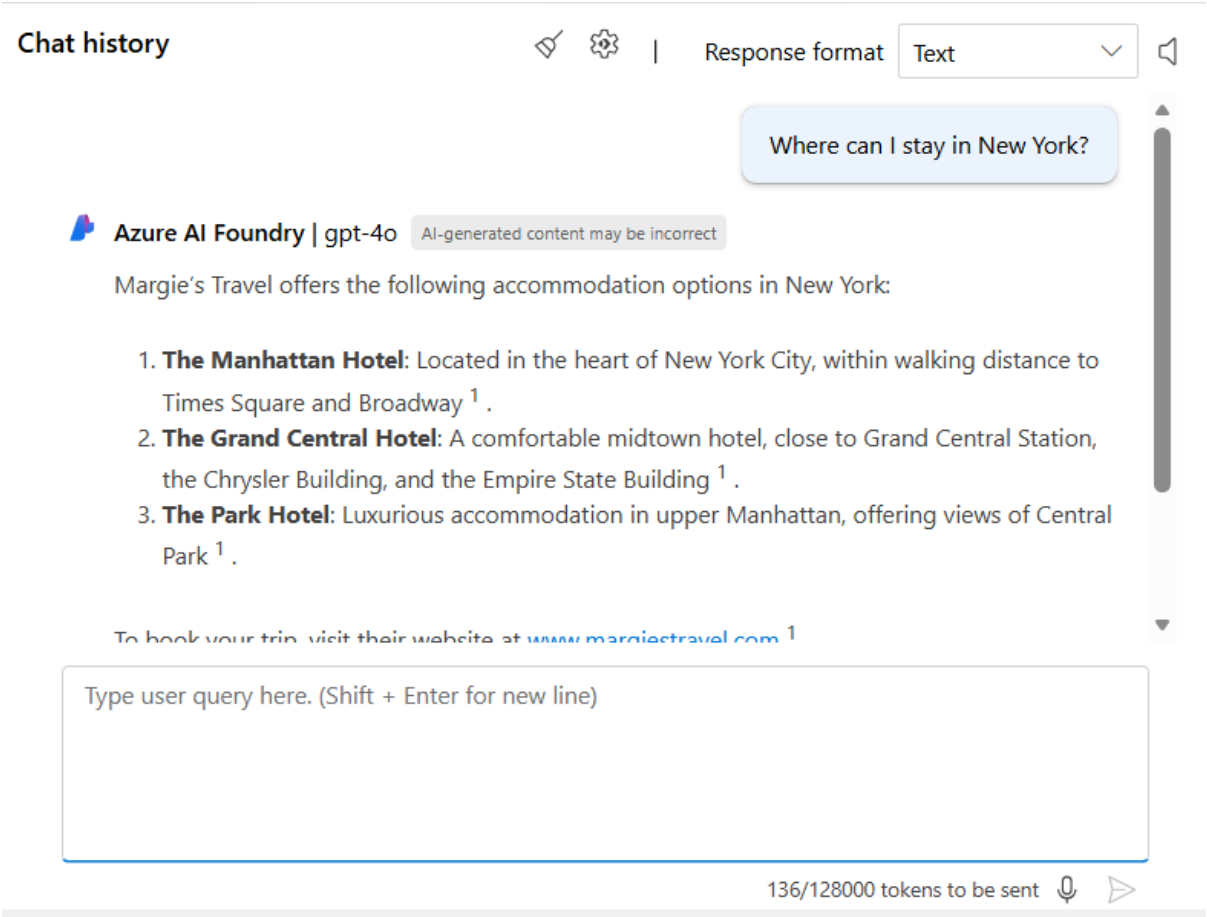
Then:

1. Add your data
2. Select brochures-index

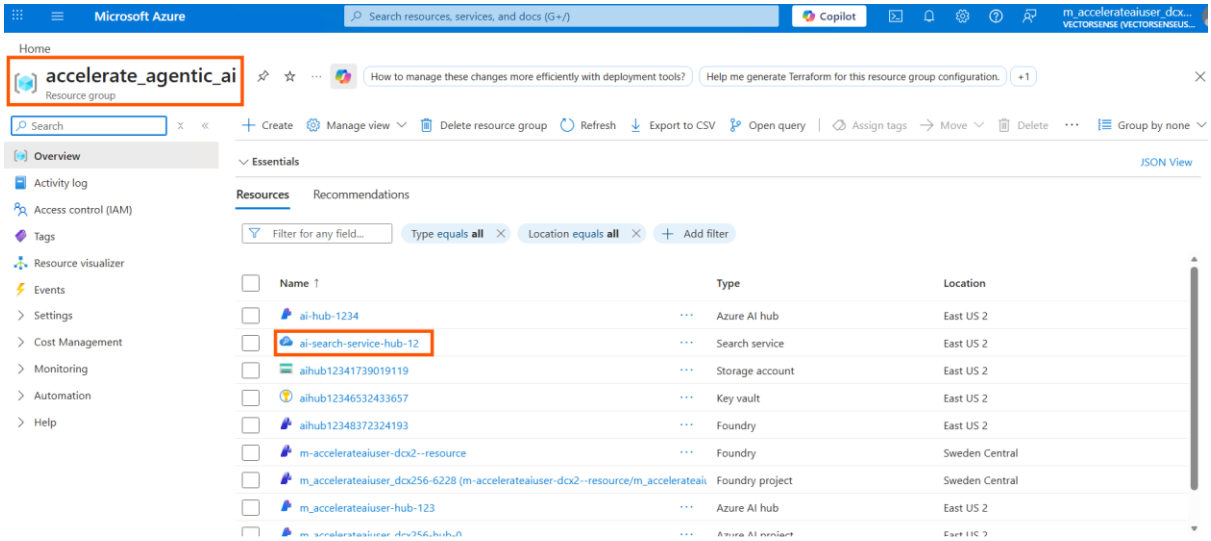


Ask again:

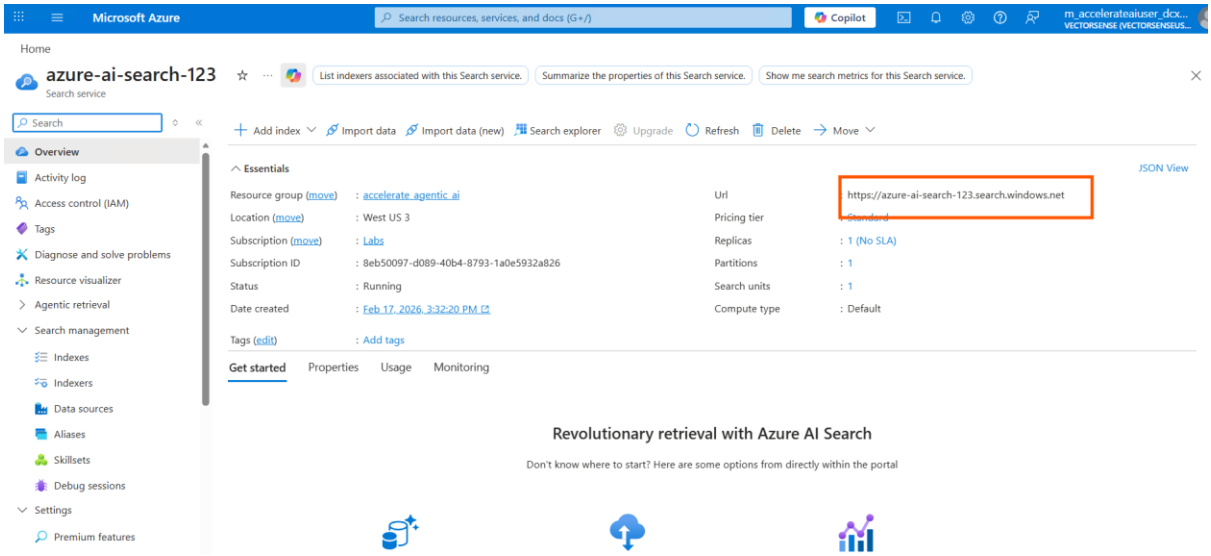
Where can I stay in New York?



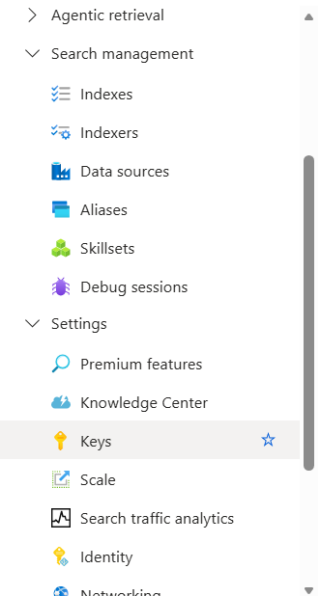
25. In a new tab open https://portal.azure.com. Next, go to **Resource group** from search bar and select **accelerate_agentic_ai** (resource group). Under the resource group, select **ai-search-service-hub-12** (search service) from there.



26. Copy the URL from the Search Service Overview and paste in notepad

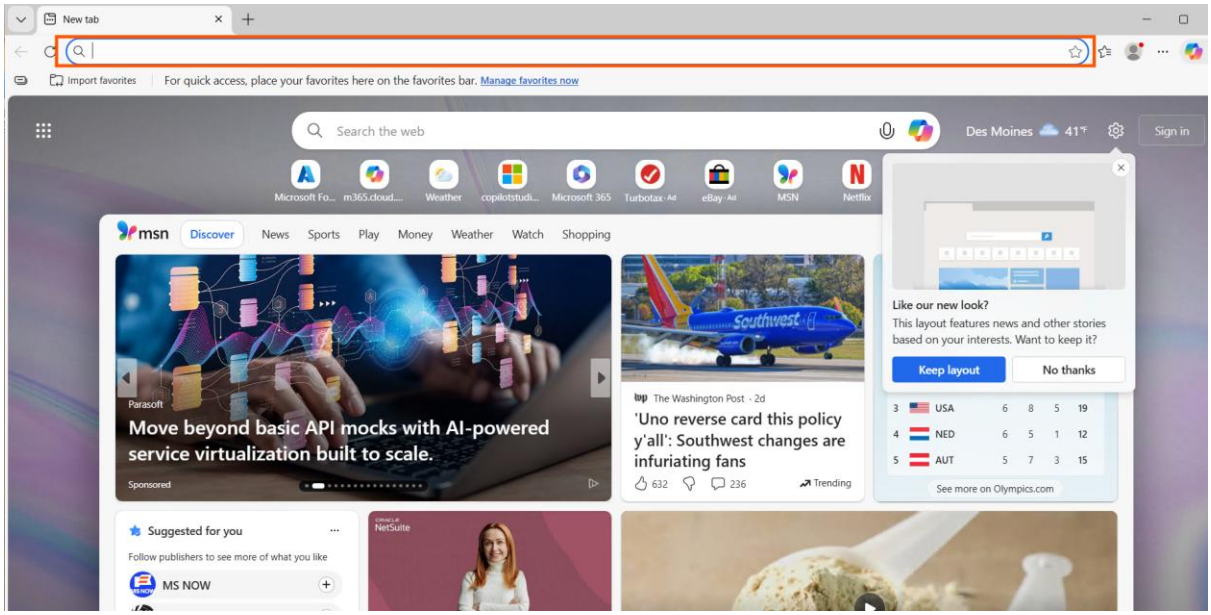


27. In the left menu pane, scroll down and click on **Keys** under Security.



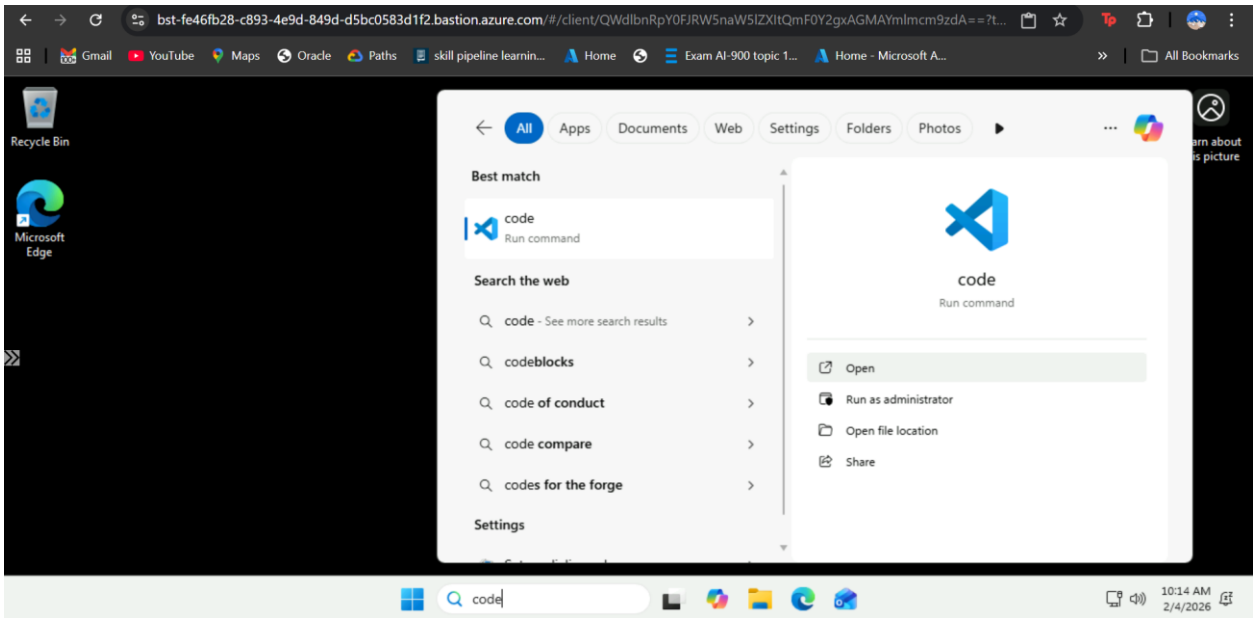
28. Click on Copy button next to the Primary admin key and note down.
29. Open a new tab and paste the below link and hit **Enter.** The required lab supporting files should automatically start downloading.

https://vectorsenselabs.blob.core.windows.net/isodata/Accelerate-Agentic_AI/Lab_1_RAG.zip?sp=r&st=2026-02-17T14:14:02Z&se=2027-02-24T22:29:02Z&spr=https&sv=2024-11-04&sr=b&sig=Rlc7ehapDllm0lyKHBBbp8EQeECFBwdRq72l5%2FMr5Ro%3D

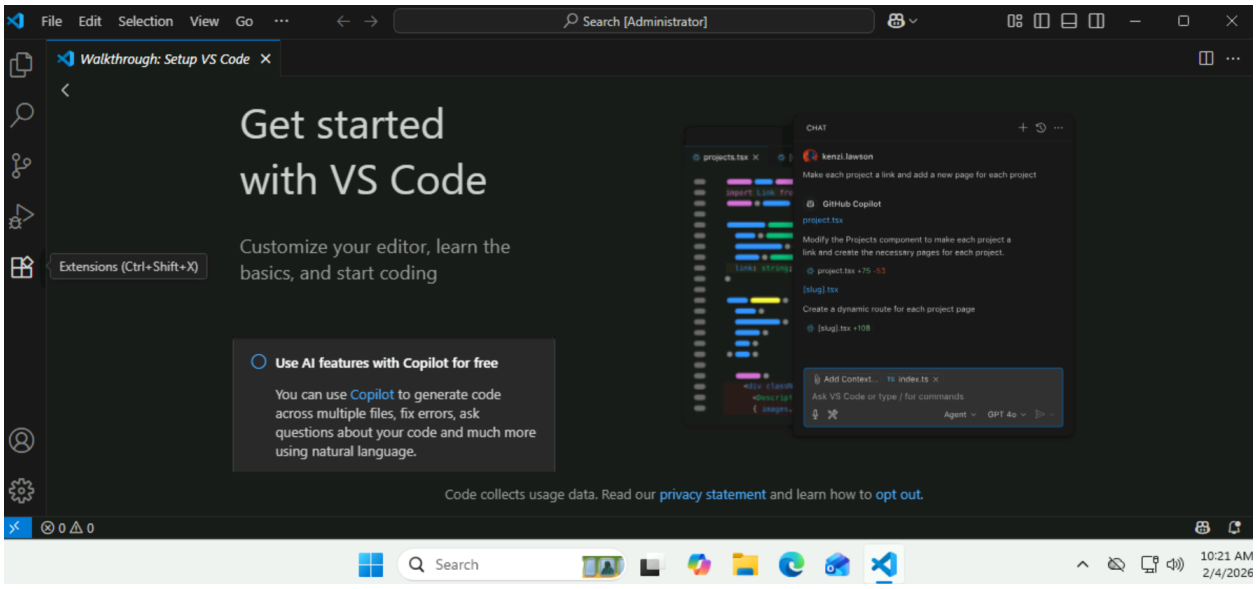


30. After the download completes, close the tab. Open **File Explorer** from the bottom menu bar. Click on **Downloads**. **Right click** over the recently downloaded folder, select **Extract All…** Click **Extract.**

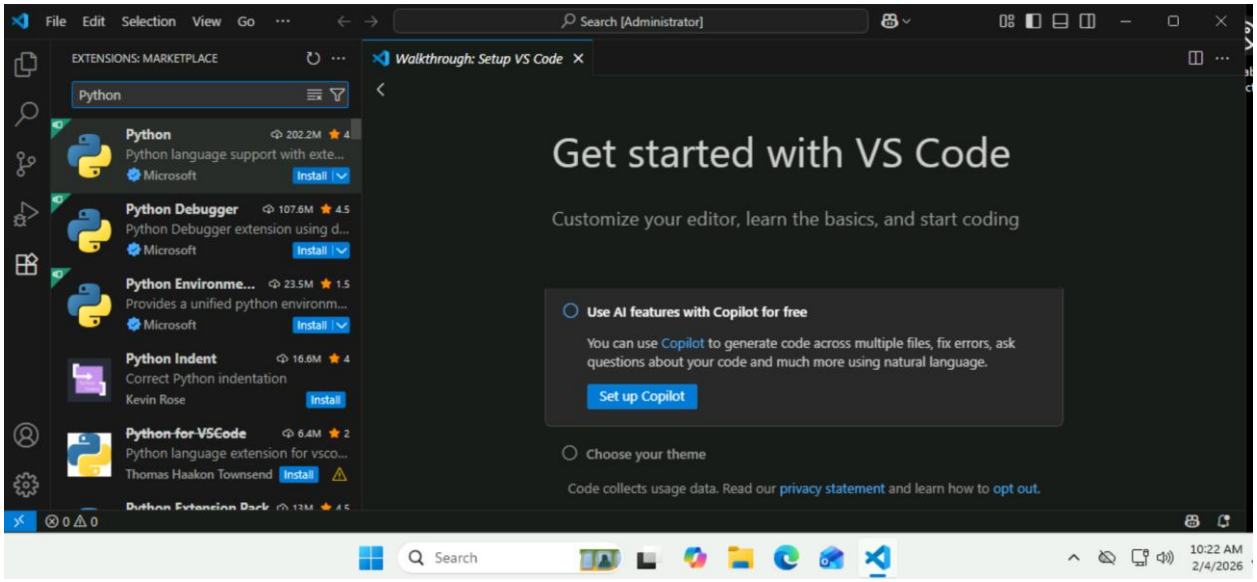
31. Close the File explorer and click on the search bar at the bottom of the screen and write **Code**. On the right side, click **Open** to launch Visual studio code.



32. Inside Visual studio code, click on **Extensions** in the left menu bar or press **Ctrl+Shift+X**



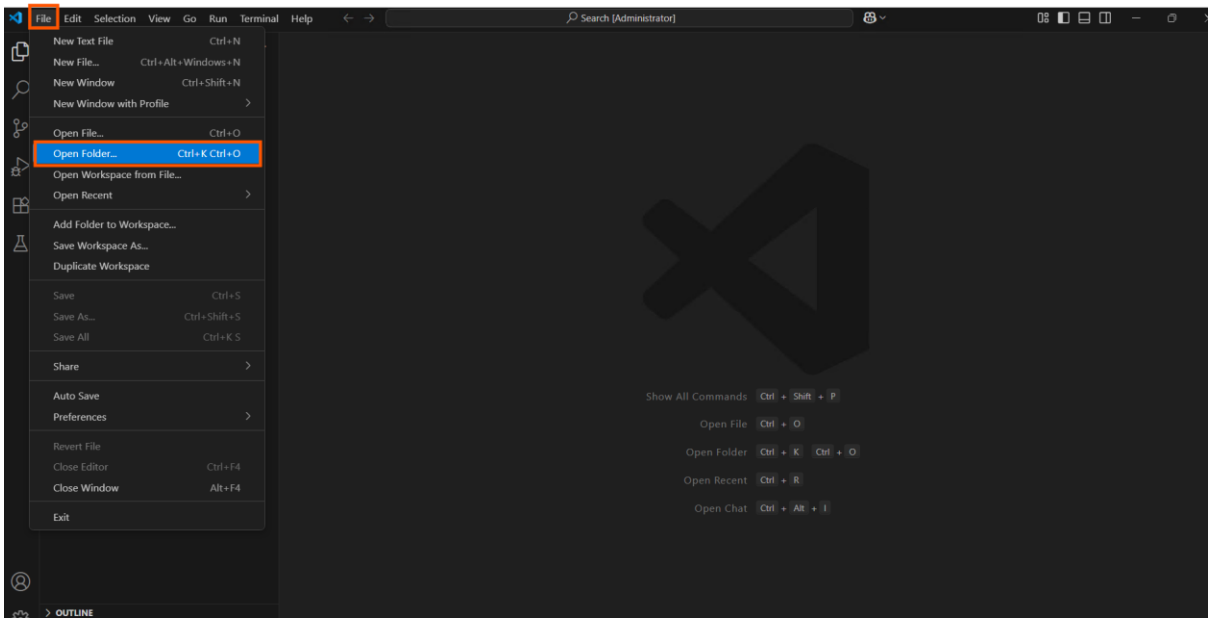
33. Search for **Python** in the text box under Extensions Marketplace. Click **Install** next to the first result. Wait for it to install before proceeding.



34. In Visual Studio Code, open the extracted lab folder:

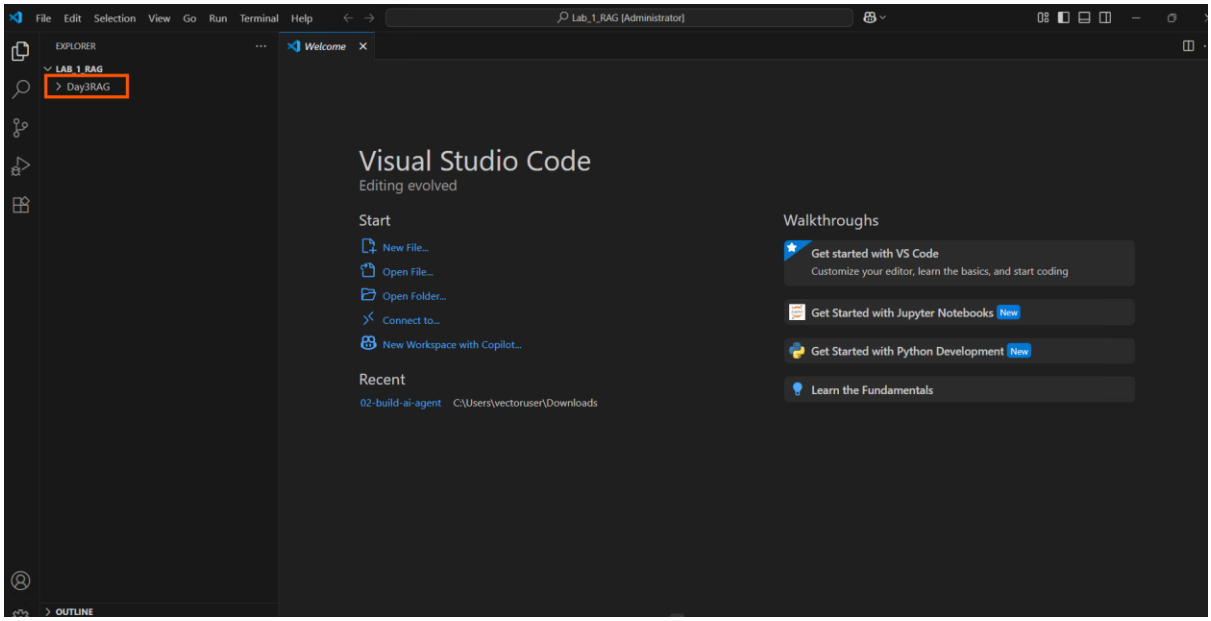
    File -> **Open Folder**

    Select the extracted **downloaded folder -> Select Folder**



    Click on **Yes, I trust the authors** if prompted.

35. Click on the **folder** in the left menu bar to view the files.

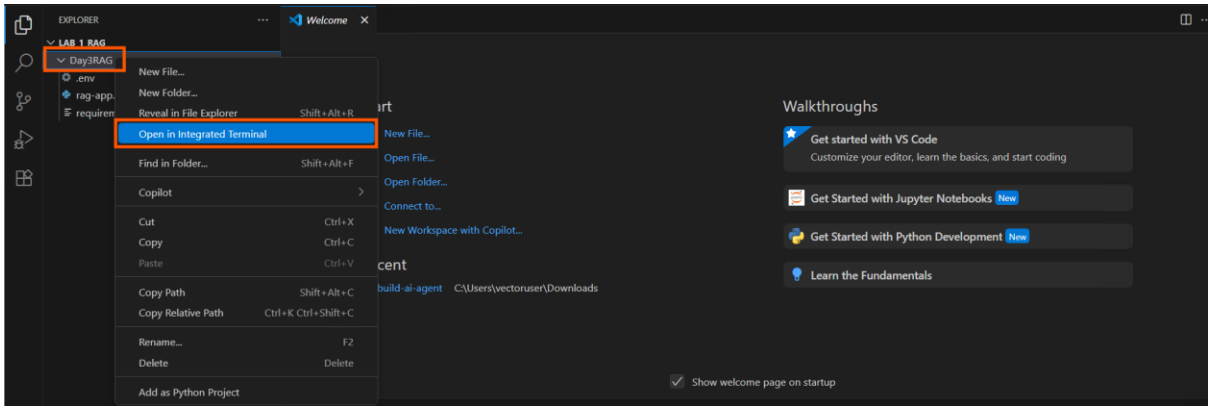


At this point, you should see files such as:

- rag-app.py
- requirements.txt
- .env

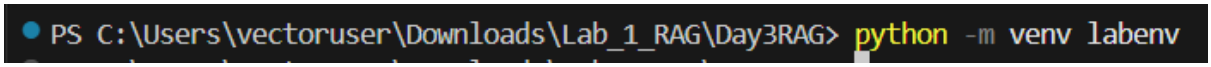
You will now create a Python environment, install required libraries, and configure the .env file so the app can connect to your Foundry project.

36. Right click on the **Day3RAG** folder from the left pane. Select **Open in Integrated Terminal.**



37. In the terminal, **run** the following commands one by one:

```
python -m venv labenv
```



```
.\labenv\Scripts\Activate.ps1
```

```
PS C:\Users\vectoruser\Downloads\Lab_1_RAG\Day3RAG> .\labenv\Scripts\Activate.ps1
```

```
pip install -r requirements.txt openai
```

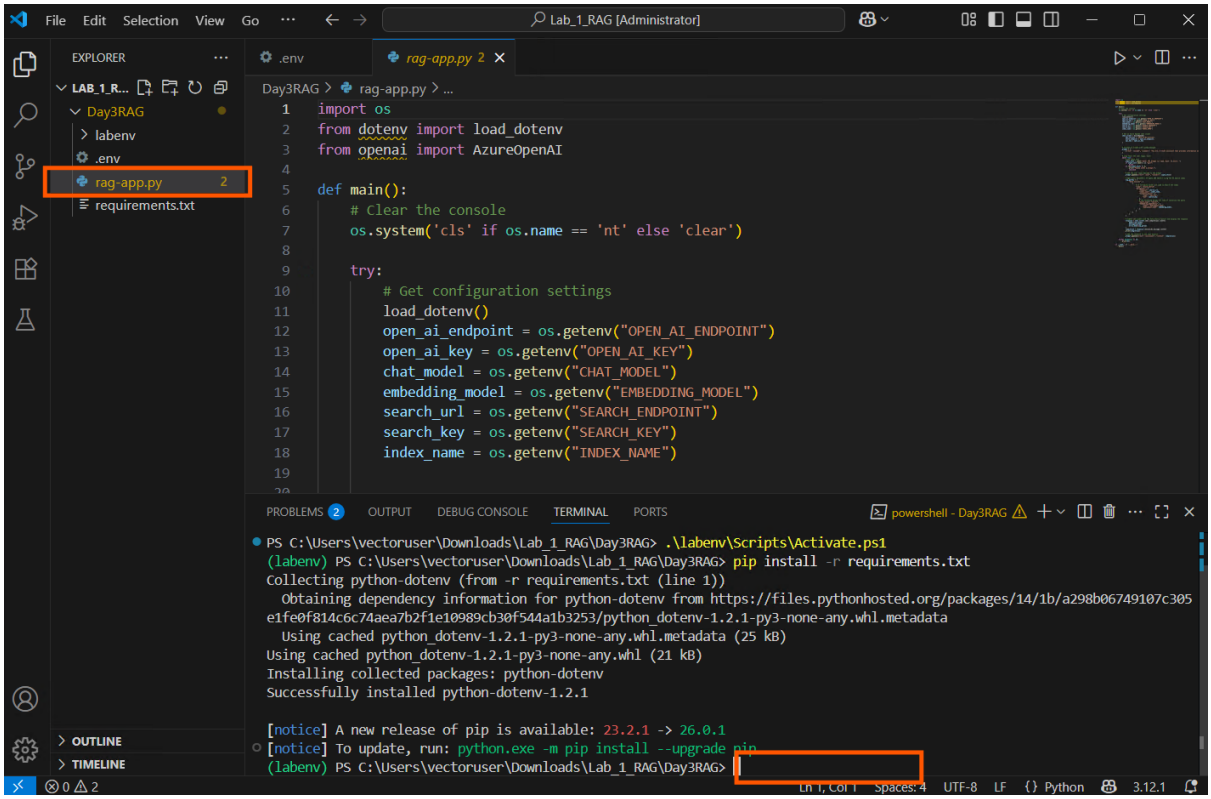
```
(labenv) PS C:\Users\vectoruser\Downloads\Lab_1_RAG\Day3RAG> pip install -r requirements.txt openai
```

38. Click twice on **.env** file to open it.
39. Update the **.env** file by replacing:

- your_openai_endpoint: Azure OpenAI endpoint from Overview → Azure OpenAI in Foundry.
- your_openai_api_key: Azure OpenAI API key from Overview → Azure OpenAI.
- your_chat_model: gpt-4o
- your_embedding_model: text-embedding-ada-002
- your_search_endpoint: Azure AI Search endpoint from Management Centre
- your_search_api_key: Azure AI Search API key
- your_index: brochures-index
  press **Ctrl + S** to save the file.

```
.env
Day3RAG > .env
OPEN_AI_ENDPOINT="https://ai-maccelerateaiuserdcx2566133a648801727860.openai.azure.com/"
OPEN_AI_KEY="FQ9WDWuLbcy8bV9uxalGXT6UPTLXEjSppy2efNZyKKxBXCjxTNk6JQQJ99CBACYeBjFXJ3w3AAAAACOGCWQk"
CHAT_MODEL="gpt-4o"
EMBEDDING_MODEL="text-embedding-ada-002"
SEARCH_ENDPOINT="https://azure-ai-search-123.search.windows.net"
SEARCH_KEY="INvj5FQjutUtbVpJPnwa6CFoQOP6cfzvfrnloZ3egZAzSeBOh7eq"
INDEX_NAME="brochures-index"
```

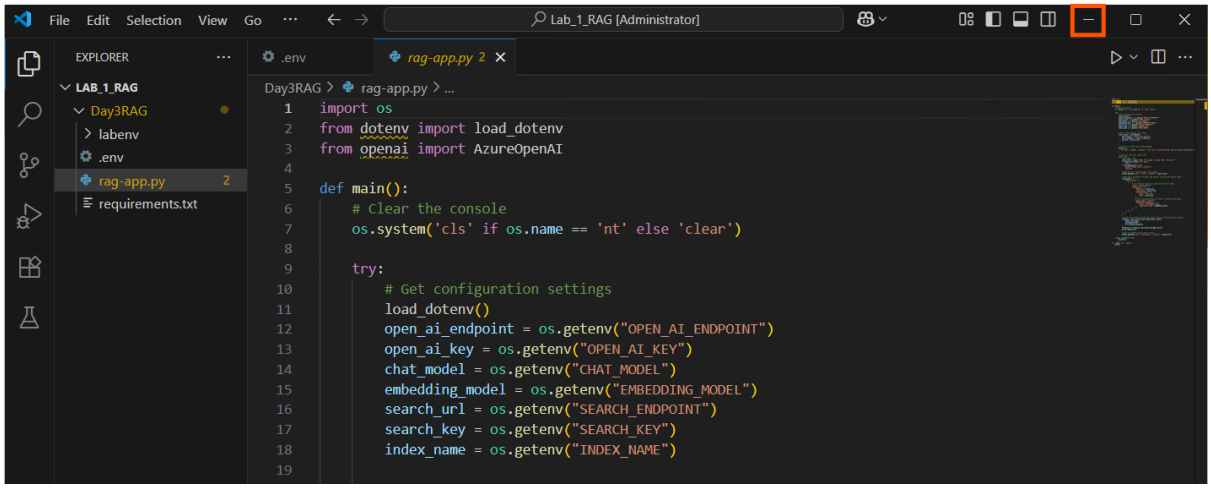
40. A Python file is already provided. Open and review it from the left pane. Click twice over the **rag-app.py** file to open it. After reviewing click on the **terminal**.



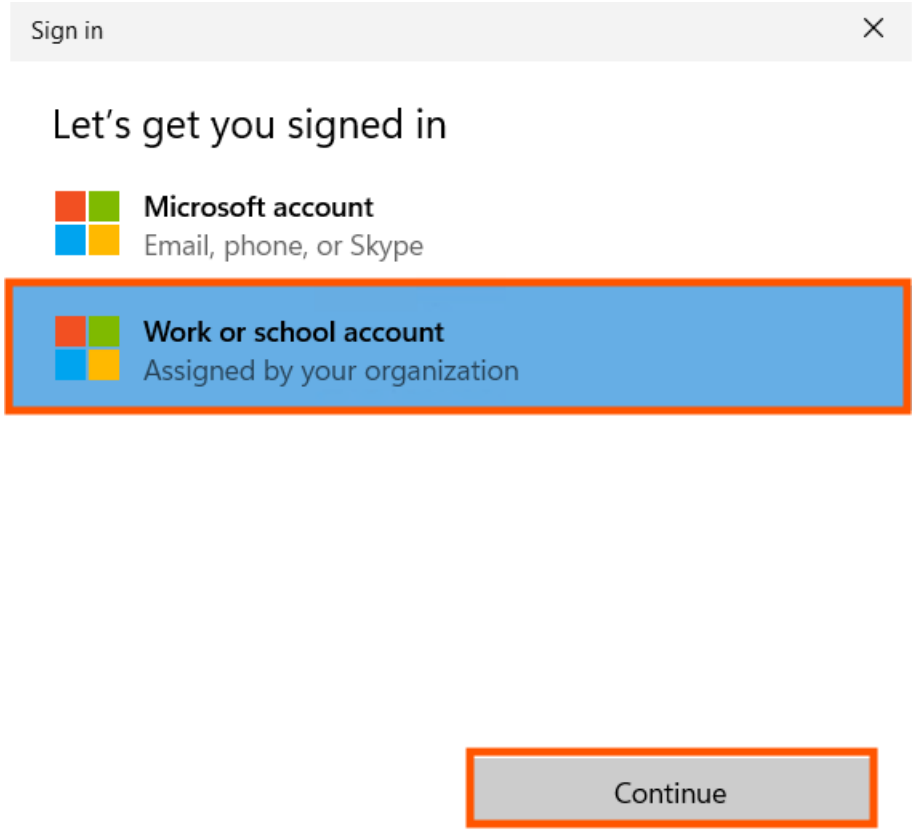
41. In terminal, **paste** the below command and hit **Enter**:

```
az login
```

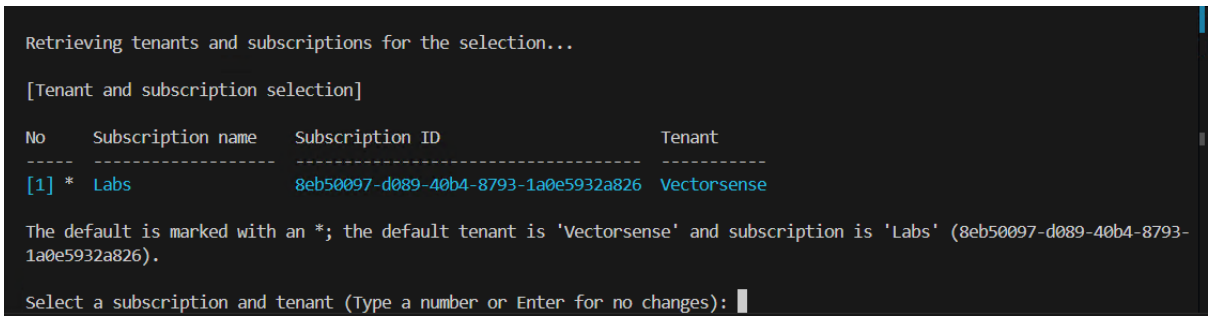
42. Click on **Minimise** button from the top left.



43. Inside **Sign In** modal, click on **Work or school account -> Continue. Sign in** with the provided details. Click on **No, this app only** if prompted.



44. Open Visual Studio Code, click on the terminal and hit **Enter.**



45. Run the following command inside the terminal:

```
python rag-app.py
```

46. Inside the terminal, write the below mentioned prompts and hit **Enter**:

Where can I stay there?

Where should I go on vacation to see architecture?

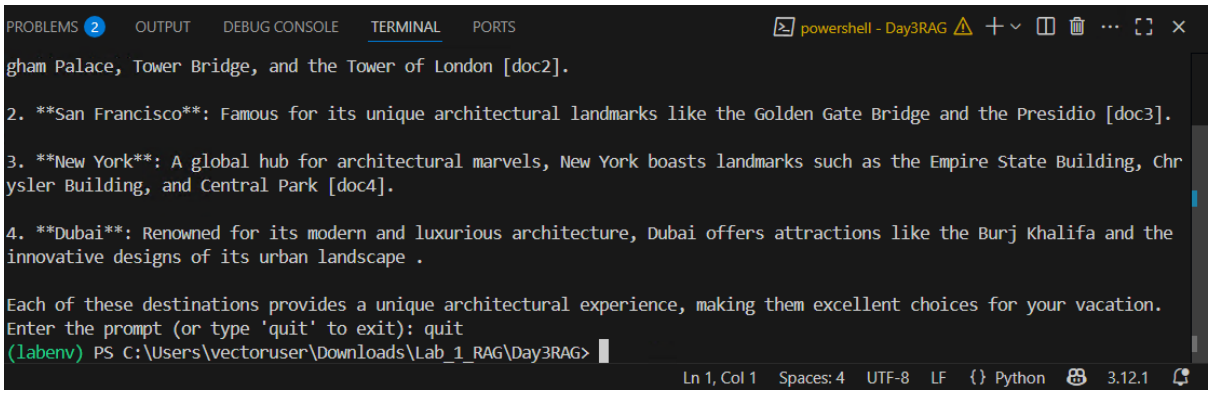
Continue or exit

You can continue asking more questions.

Type:

quit

…when you are done



47. Write `deactivate` in the terminal and hit **Enter**, the virtual environment will be deactivated.

Lab Summary

In this lab, you:

- Created a Foundry hub and project
- Deployed text-embedding-ada-002 and gpt-4o
- Uploaded PDF data
- Created a vector index
- Tested RAG in Chat Playground
- Built and ran a Python RAG client in Visual Studio Code

Congratulations, you have completed the Lab. Please close the current document and move to the next one.